

Yuni Wu

South San Francisco, CA

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Summary

Robotics AI Engineer focused on making robot behavior measurable, reproducible, and reliable. Builds evaluation datasets and benchmarks that help teams understand robot decisions in real-world environments.

Technical Skills

Languages: Python, SQL, Go, R

Robotics Data: ROS, rosbag workflows, Unitree Go1 trajectories, telemetry processing, state-action datasets, scenario analysis

Navigation AI: VLM navigation, behavior cloning, semantic scene understanding, embodied AI, robot decision analysis

Evaluation: offline evaluation, scenario benchmarking, action consistency, deployment robustness, inference latency analysis

ML & Systems: ResNet, Transformers, calibration, cross-validation, Docker, Cloud Run, validation workflows, metadata pipelines

Experience

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Engineered ROS/rosbag data pipelines to curate 97K+ state-action samples from Unitree Go1 trajectories, supporting imitation-learning and embodied-AI research.
- Benchmarked ResNet-based behavior-cloning policies for autonomous navigation across diverse human-interaction and indoor navigation scenarios.
- Analyzed VLM-guided navigation strategies to evaluate how semantic scene understanding influences robot decision-making, action consistency, and deployment robustness.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Built a 22-class classification workflow on 50K+ samples, achieving 0.92 F1-score through repeatable evaluation.
- Evaluated deployment tradeoffs through ablation workflows, reducing inference latency by 5x from 0.15s to 0.03s per image.
- Designed cross-validation, ROC-AUC, calibration, monitoring, and regression validation workflows to measure reliability.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered SQL validation and anomaly-detection pipelines for 800+ production scenarios, reducing manual QA by 80%.
- Built regression monitoring pipelines to surface data inconsistencies faster, accelerating reporting speed by 78%.
- Optimized distributed ETL workflows and GCP data-access patterns, reducing query latency by 15%.

Projects

VLM-Based Social Navigation for Quadruped

[project]

- Deployed and evaluated VLM-based social-navigation systems on a Unitree Go1 quadruped, comparing InternVL, Qwen-VL, and prompt-engineering strategies for human-aware navigation.
- Constructed and benchmarked an offline evaluation dataset containing 2.4K+ labeled state-action sequences across 13 robot-navigation scenarios, including hallway traversal and intersection crossing.
- Assessed VLM decision-making consistency across scenario-level navigation tasks to study human-aware action selection.
- Trained and deployed ResNet-based behavior-cloning policies on 97K+ state-action samples, achieving 0.11 MAE and $R^2 = 0.62$ in waypoint prediction for autonomous hallway navigation.

NailAI — Cloud-Native AI Processing Platform

[demo]

- Enabled fault-tolerant asynchronous image processing through Cloud Run services, Pub/Sub job routing, and Cloud Storage workflows.
- Achieved 0% failure at 200 concurrent users with p95 latency of 6.8s through idempotent job orchestration and retries.
- Designed metadata ingestion pipelines for request, job, and prediction tracking using BigQuery-backed analytics and monitoring workflows.
- Optimized BigQuery partitioning and clustering for warehouse-centric downstream analysis and reduced query latency.

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0

May 2026

Soochow University — BS, Data Science | GPA: 3.7/4.0

June 2021